









Time Series Analysis

Seminar 12: Spectral analysis



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BIC**: Bayesian Information Criterion
- **DFT**: Discrete Fourier Transform
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HP**: Hodrick–Prescott (filter)
- **SD**: Standard Deviation
- **SILSO**: Sunspot Index and Long-term Solar Observations
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)

Today's Question and Seminar Structure

- The question of the seminar
  - **which cycles does a series contain, how strong are they, and which ones do our filters keep or remove?**
- Results to obtain (practice only: nothing is handed in or graded)
  - the answers to the A exercises, computed by hand
  - a figure and a table for each B exercise
  - the corrected AI answer of C1
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	frequencies, periods and aliasing; the periodogram of four values	Solved, Proposed	A1
A3, A4	the AR(1) spectrum of US GDP growth; the AR(2) spectrum of the sunspots	Solved, Proposed	A3
A5	gains of differences and of the Hodrick–Prescott (HP) filter	Solved	A5
B1, B2	daily and weekly cycles of the Romanian load; the business cycle in US industrial production	Solved, Proposed	B1
C1	what is wrong in an AI answer about spectra?	Proposed	A1–B2

- Types: **A** calculations on paper; **B** real data, in the notebook; **C** an AI critique
- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar

Data used

Series	Source	Frequency	Period
electricity load, Romania	ENTSO-E	hourly (UTC)	2025
real GDP growth, US	FRED (GDPC1)	quarterly	1947–2019
yearly sunspot number	SILSO (statsmodels)	yearly	1700–2008
industrial production growth, US	FRED (INDPRO)	monthly	1948–2019

- Growth rates: $100\Delta \ln x_t$; GDP and industrial production end in 2019, before the 2020 outlier, as in Lecture 12
- In the notebook: `periodogram`, `daniell`, `chi2_band`, `arma_spectrum`, `hp_gain`, `fit_ar`; no account or key is needed

Prerequisites for Today (1/5): frequency, period and aliasing

- **Frequency** ν: cycles per observation; **period** $1/\nu$: observations per cycle Shumway and Stoffer (2017)
  - monthly data, a yearly cycle: $\nu = 1/12$; hourly data, a daily cycle: $\nu = 1/24$
- **Fourier frequencies** of T observations: $\nu_j = j/T$, $j = 1, \ldots, \lfloor T/2 \rfloor$
  - a cycle of period p sits exactly on a Fourier frequency only if T/p is an integer
- **Nyquist frequency** $\nu = 1/2$: the fastest visible cycle has a period of 2 observations
- **Aliasing**: a cycle with $\nu > 1/2$ looks like a slower one, at the apparent frequency

$$\nu_a = |\nu - \text{round}(\nu)| \in [0, 1/2]$$

  - with ν measured in cycles per *sampling interval*

Prerequisites for Today (2/5): the periodogram

- **Discrete Fourier transform** (DFT) of the demeaned series $y_t = x_t - \bar{x}$

$$d(\nu_j) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} y_t e^{-2\pi i \nu_j t}, \qquad I(\nu_j) = |d(\nu_j)|^2$$

  - $e^{-i\theta} = \cos\theta - i \sin\theta$; $|a + bi|^2 = a^2 + b^2$

- **Parseval**: the periodogram splits the sum of squares over the frequencies

$$\sum_{j=1}^{T-1} I(\nu_j) = \sum_{t=1}^{T} y_t^2$$

  - $I(\nu_j) = I(1 - \nu_j)$: each frequency below $1/2$ appears twice
  - share of a band: the sum of $I(\nu_j)$ in the band over the sum of all $I(\nu_j)$, $0 < \nu_j \leq 1/2$

Prerequisites for Today (3/5): spectral densities of ARMA models

- **Spectral density** $f(\nu) = \sum_h \gamma(h) e^{-2\pi i \nu h}$; $\int_{-1/2}^{1/2} f(\nu)\, d\nu = \gamma(0)$, the variance
  - white noise: $f(\nu) = \sigma^2$, flat
- **AR(1)** $x_t = \phi x_{t-1} + w_t$

$$f(\nu) = \frac{\sigma^2}{1 - 2\phi\cos(2\pi\nu) + \phi^2}$$

  - $\phi > 0$: most power at low frequencies; no interior peak
- **AR(2)** with complex roots ($\phi_1^2 + 4\phi_2 < 0$): a pseudo-cycle

$$\cos(2\pi\nu^*) = \frac{\phi_1(\phi_2 - 1)}{4\phi_2}, \quad \text{a peak inside } (0, 1/2) \text{ only if this value is in } (-1, 1)$$

Prerequisites for Today (4/5): smoothing and confidence bands

- **The raw periodogram is inconsistent**: $2I(\nu_j)/f(\nu_j) \approx \chi^2_2$ for every T, so its noise never shrinks
- **Daniell smoother**: the mean of $L = 2m + 1$ neighbouring ordinates

$$\hat{f}(\nu_j) = \frac{1}{L} \sum_{k=-m}^{m} I(\nu_{j+k}), \qquad \text{df} = 2L, \qquad \text{bandwidth } B = L/T$$

  - larger L: less noise, but peaks closer than B merge
- **95% confidence interval** for $f(\nu)$, from $\text{df}\,\hat{f}/f \approx \chi^2_{\text{df}}$

$$\left[\frac{\text{df}\,\hat{f}}{\chi^2_{\text{df}}(0.975)}, \; \frac{\text{df}\,\hat{f}}{\chi^2_{\text{df}}(0.025)}\right]$$

  - not symmetric around $\hat{f}$; on a log scale its width is the same at every frequency

Prerequisites for Today (5/5): linear filters and their gains

- **Filter** $y_t = \sum_k a_k x_{t-k}$: $f_y(\nu) = |A(\nu)|^2 f_x(\nu)$, with $A(\nu) = \sum_k a_k e^{-2\pi i \nu k}$; $|A(\nu)|^2$: the squared gain
- **Differences**
  - $1 - L$: $|A(\nu)|^2 = 4\sin^2(\pi\nu)$, zero at $\nu = 0$; $1 - L^s$: $4\sin^2(\pi s\nu)$, zero at $\nu = k/s$
- **Hodrick–Prescott (HP) cycle** Hodrick and Prescott (1997): gain with $\omega = 2\pi\nu$

$$G(\nu) = \frac{4\lambda(1 - \cos\omega)^2}{1 + 4\lambda(1 - \cos\omega)^2}$$

  - $G \approx 1$ for short cycles (kept in the cycle), $G \approx 0$ for long ones (sent to the trend); the trend filter has gain $1 - G$

A1: frequencies, periods and aliasing [Solved]

Task

- One year of hourly electricity load: $T = 365 \cdot 24 = 8760$ hours.
- 1. Find the Fourier index j of the 24-hour, 12-hour and 168-hour cycles. Is the weekly cycle a Fourier frequency?
- 2. What is the shortest period the hourly data can show?
- 3. A meter is read once every 24 hours, every 8 hours or every 25 hours. Where do the daily and the 12-hour cycles appear?
- Report: three indices, one period, three apparent frequencies.

Solution

- 1. $j = T/p$: $8760/24 = 365$, $8760/12 = 730$, $8760/168 = 52.14$: not an integer; nearest $j = 52$, period 168.46 hours (leakage to the neighbours)
- 2. Nyquist $\nu = 1/2$: a period of 2 hours
- 3. Daily cycle read every 24 h: $\nu = 24/24 = 1.000$, $\nu_a = 0.000$: it vanishes into the level
- 12-hour cycle read every 8 h: $\nu = 8/12 = 0.667$, $\nu_a = 0.333$: period 3 readings = 24 hours, a false daily cycle
- Daily cycle read every 25 h: $\nu = 25/24 = 1.0417$, $\nu_a = 0.0417$: a slow false cycle of 600 hours (25 days)

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A2: the periodogram of four values [Proposed]

Solution discussed in class

Task

- The series $x = (2, 6, 4, 0)$, $t = 1, \ldots, 4$. Model: A1.
- 1. Remove the mean and list the Fourier frequencies $\nu_j = j/4$, $j = 1, 2, 3$.
- 2. Compute $d(\nu_j)$ and $I(\nu_j)$ (use $e^{-i\pi t/2} = -i, -1, i, 1$ for $t = 1, \ldots, 4$).
- 3. Check Parseval and say which period carries the whole variance.
- Report: three ordinates, one check, one period.

A3: the AR(1) spectrum of US GDP growth [Solved]

Task

- US real GDP growth, quarterly, 1947–2019 ($T = 291$); an AR(1) by least squares: $\hat{\phi} = 0.358$, $\hat{\sigma}^2 = 0.752$.
- 1. Compute $f(0)$, $f(1/4)$, $f(1/2)$ and the ratio $f(0)/f(1/2)$.
- 2. Compute $\gamma(0) = \sigma^2/(1-\phi^2)$ and compare it with the sample variance.
- 3. The share of the variance at periods of 6 to 32 quarters is 43.3% for this AR(1) and 48.8% in the periodogram. What would white noise give?
- Report: three values, one ratio, one variance, one share.

Solution

- 1. $\cos 0 = 1$: $f(0) = \hat{\sigma}^2/(1-\hat{\phi})^2 = 1.827$; $\cos(\pi/2) = 0$: $f(1/4) = \hat{\sigma}^2/(1+\hat{\phi}^2) = 0.667$; $\cos\pi = -1$: $f(1/2) = \hat{\sigma}^2/(1+\hat{\phi})^2 = 0.408$
- ratio $\left((1+\hat{\phi})/(1-\hat{\phi})\right)^2 = 4.48$
- 2. $\gamma(0) = 0.863$, sample variance 0.862
- 3. White noise: the band $[1/32, 1/6]$ is 27.1% of $[0, 1/2]$
- the AR(1) shifts power to low frequencies but has no peak; the data put a bit more variance on business-cycle periods than the AR(1)

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A4: the AR(2) spectrum of the sunspots [Proposed]

Solution discussed in class

Task

- Yearly sunspot numbers 1700–2008 ($T = 309$); an AR(2) by least squares: $\hat{\phi}_1 = 1.392$, $\hat{\phi}_2 = -0.690$. Model: A3.
- 1. Show that the roots are complex.
- 2. Compute the peak frequency ν^* of the spectrum and its period in years.
- 3. The ACF of an AR(2) oscillates with period $2\pi / \arccos(\phi_1/(2\sqrt{-\phi_2}))$. Compute it and compare both with the largest periodogram ordinate, $j = 28$.
- Report: one discriminant, one frequency, three periods.

A5: what do differences and the HP filter keep? [Solved]

Task

- Quarterly data; first difference $1 - L$ and the HP cycle filter with $\lambda = 1600$.
- 1. Compute the squared gain of $1 - L$ at periods 2, 4, 8 and 32 quarters.
- 2. Compute G^2 of the HP cycle at periods 8, 32 and 40 quarters.
- 3. Find the period where $G^2 = 1/2$ for $\lambda = 1600$ (quarterly), 129 600 (monthly) and 6.25 (annual).
- Report: four gains, three gains, three periods in years.

Solution

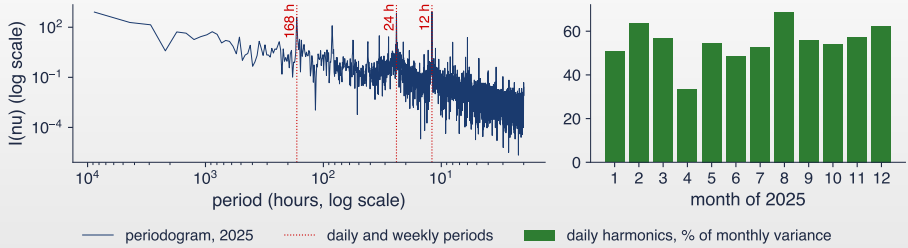
- 1. $4\sin^2(\pi/p)$: 4.000, 2.000, 0.586, 0.038: differencing amplifies short cycles and nearly removes long ones
- 2. G^2: 0.996 (8 quarters), 0.494 (32), 0.242 (40): the cut-off is gradual
- 3. $G^2 = 1/2 \Leftrightarrow 4\lambda(1 - \cos\omega)^2 = 1/(\sqrt{2} - 1) = 2.414$, so $\cos\omega = 1 - \sqrt{2.414/(4\lambda)}$
- $\lambda = 1600$: $\cos\omega = 0.9806$, 31.8 quarters = 7.96 years; 129 600: 95.6 months = 7.97 years; 6.25: 7.75 years
- the three usual values of λ put the cut-off near 8 years

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B1: cycles of the Romanian electricity load [Solved]

- **Question**: how much of the hourly load is explained by the daily and the weekly cycle, and is the daily cycle equally strong all year?
- **Inputs**: hourly load of Romania in 2025 (ENTSO-E, UTC hours, $T = 8760$, mean 6.11 GW)
- **Tasks**
  1. Compute the periodogram and list the six largest peaks with distinct periods, in hours and in days.
  2. Compute the share of the variance at the daily harmonics (24, 12, 8, 6, 4.8, 4 hours) and at the weekly ones (± 2 Fourier frequencies).
  3. Repeat the daily share for each month of 2025.
  4. Interpretation: why is the 12-hour peak as large as the 24-hour one, and why does the daily share change over the year?
- **Report**: a table of six peaks, two shares, two months, one sentence
- Notebook: section B1

B1: solution [Solved]



- Daily harmonics 39.0% of the variance, weekly 16.3%; the daily share is largest in August (69%) and smallest in April (33%)
- Interpretation: the day has two load peaks (morning, evening), so the profile needs the 12-hour harmonic as much as the 24-hour one; summer air conditioning and winter lighting strengthen the daily cycle, mild April weather weakens it

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B1: the largest peaks [Solved]

#	period (hours)	period (days)	share of variance (%)
1	12.0	0.50	18.0
2	8760.0	365.00	17.5
3	24.0	1.00	15.1
4	168.5	7.02	8.6
5	4380.0	182.50	4.2
6	2920.0	121.67	3.0

- The largest ordinate is the year itself (18.0%): the seasonal swing of the level, which one year of data sees as a single slow cycle
- The weekly cycle appears at $j = 52$, period 168.46 hours, the Fourier frequency closest to 168 hours (A1)

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B2: the business cycle in US industrial production [Proposed]

- **Question**: is there a significant peak at business-cycle periods, and does a low-order AR spectrum find it?
- **Inputs**: monthly growth of US industrial production, February 1948 – December 2019 ($T = 863$); model: B1
- **Tasks**
  1. Compute the periodogram and smooth it with a Daniell window of $L = 7$; give the bandwidth and the degrees of freedom.
  2. Find the peak of $\hat{f}$ at periods of 18–96 months and its 95% interval; compare with the level of $\hat{f}$ at periods below 6 months.
  3. Compute the share of the variance in the 18–96-month band and compare with white noise.
  4. Fit AR(p), $p \leq 24$, choose p by BIC and plot its spectrum. Interpretation: why does it miss the peak?
- **Report**: one bandwidth, one interval, two shares, one AR order, one sentence
- Notebook: section B2

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C1: audit an AI answer [Proposed]

- **A student asked an AI assistant about spectral analysis. The answer:**
- `(a) The sunspot periodogram peaks at nu = 0.09, so the solar cycle lasts 0.09 years.`
- `(b) With more data the raw periodogram converges to the true spectrum.`
- `(c) Daily data cannot show a weekly cycle, because the Nyquist frequency is 0.5.`
- `(d) Every AR(2) with phi2 < 0 has a spectral peak inside (0, 1/2).`
- `(e) The HP filter with lambda = 1600 removes every fluctuation longer than 8 years.`
- `(f) A 95% Daniell interval with L = 5 is the estimate plus or minus 1.96 standard errors.`
- **Tasks**
  1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (A1–B2, notebook section C1).
- **Report**: a list of six verdicts with one line of justification each

Key takeaways

- Period $= 1/\nu$ in the units of the sampling interval; cycles faster than the Nyquist frequency reappear as slower ones
- The periodogram splits the variance over frequencies (Parseval) but must be smoothed; Daniell intervals come from χ^2_{2L}
- AR(1): power at low frequencies; AR(2) with complex roots: a peak, as for the 11-year solar cycle
- Filters act frequency by frequency: differences remove the long cycles, HP has a gradual cut-off near 8 years
- An AI answer is a draft: check units, the sampling interval and the distribution of the estimate

After the seminar

- Lecture 12 gives the details
  - leakage and tapering, Welch estimates, Fisher's test, long memory, coherence and phase, wavelets
- Try the [Proposed] tasks in the notebook
- Project idea: the spectrum of Romanian industrial production or of the hourly load in other countries
- Reading: Shumway and Stoffer (2017), Ch. 4; Hodrick and Prescott (1997)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- Bartlett, M. S. (1950). Periodogram analysis and continuous spectra. *Biometrika*, 37(1–2), 1–16. doi:10.1093/biomet/37.1-2.1
- Brockwell, P. J., & Davis, R. A. (1991). *Time Series: Theory and Methods* (2nd ed.). Springer. doi:10.1007/978-1-4419-0320-4
- Hamilton, J. D. (2018). Why you should never use the Hodrick–Prescott filter. *Review of Economics and Statistics*, 100(5), 831–843. doi:10.1162/rest_a_00706
- Hodrick, R. J., & Prescott, E. C. (1997). Postwar U.S. business cycles: An empirical investigation. *Journal of Money, Credit and Banking*, 29(1), 1–16. doi:10.2307/2953682
- Shumway, R. H., & Stoffer, D. S. (2017). *Time Series Analysis and Its Applications: With R Examples* (4th ed.). Springer. doi:10.1007/978-3-319-52452-8